

Trainings ▾

Preparatory Steps

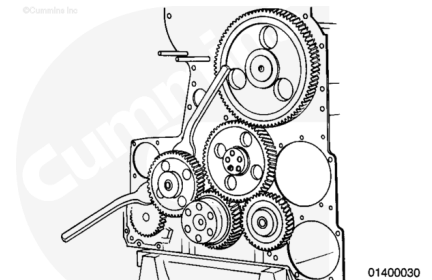
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html>)

Remove

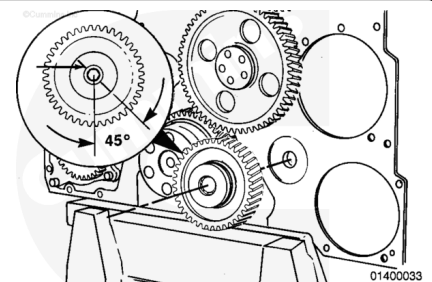
Note : The bolt-in idler shafts have flanges that require the shaft, the idler gear, and the thrust washer to be removed as an assembly.

Remove the capscrew from the idler gear.

Note : To remove the gear assembly, use two pry bars and pry the gear and the shaft from the block.



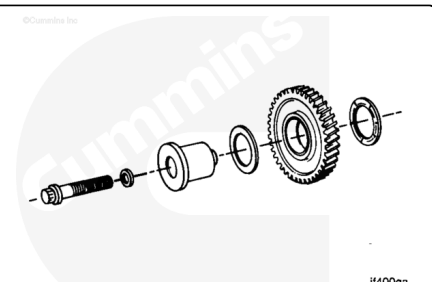
Remove the hydraulic pump idler shaft and thrust bearings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

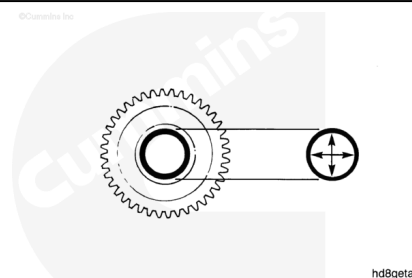
Use solvent. Clean the idler gear.

Dry with compressed air

Check all parts for damage.

Measure the inside diameter of the idler.

Idler Gear Inside Diameter		
mm		in
47.638	MIN	1.8755
47.663	MAX	1.8765



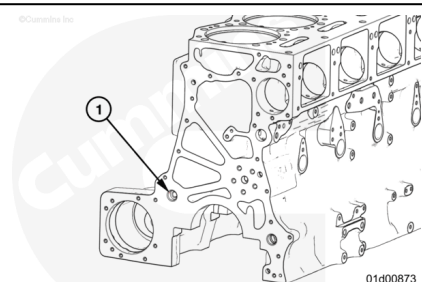
Measure the outside diameter of the idler.

Idler Gear Bushing Outside Diameter (1)		
mm		in
47.587	MIN	1.8735
47.600	MAX	1.8740

Note : The bushing in the gear is precision bored after installation. If machining capability is **not** available, replace the bushing and the gear as an assembly.

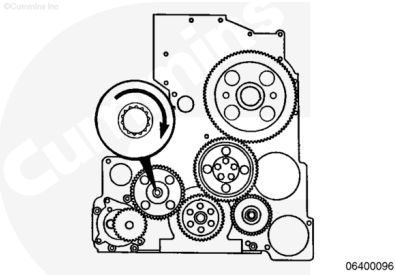
Measure the water pump idler block bore diameter.

Water Pump Idler Block Bore Diameter (1)		
mm		in
25.35	MIN	0.998
25.40	MAX	1.000



Install

Install the water pump idler gear.

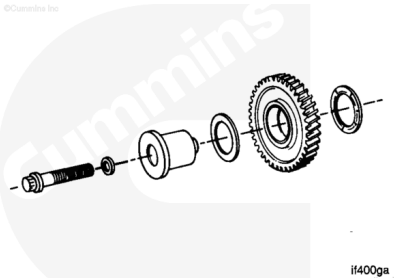


⚠ CAUTION ⚠

The grooves in the thrust washers must be turned toward the gear.

Use Lubriplate™ 105 multipurpose lubricant, or equivalent. Lubricate the gear bushing, shaft, and the thrust washer.

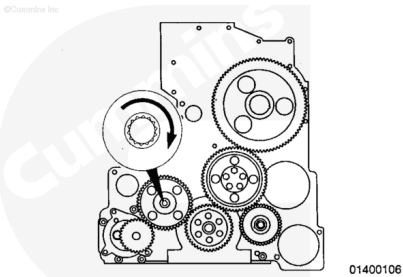
Use engine oil. Lubricate the capscrew. Assemble the parts as shown.



Align the oil holes in the shaft to an angle 45 degrees to the left of vertical.

Use the capscrew to pull the shaft into the bore. Tighten the capscrews.

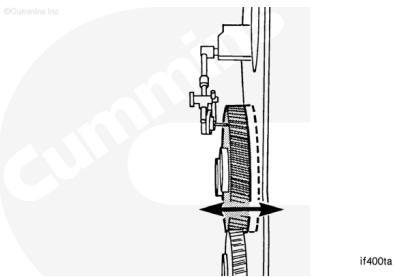
Torque Value: 245 n•m [180 ft-lb]



Use a dial indicator and measure the idler gear end clearance.

Water Pump - Idler Gear End Clearance		
mm		in
0.10	MIN	0.004
0.36	MAX	0.014

Note : If the clearance is **not** within specifications, check for foreign material between the parts, or check for proper location of the thrust washers. Oversize washers are available.



Finishing Steps

- Install the front gear cover. Refer to Procedure 001-031 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)

Last Modified: 16-Apr-2019
